Priyadharshini Sivakumar

Artificial Intelligence Engineer (LLMs & Generative AI)

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SUMMARY

AI Engineer with 4+ years building production-grade Generative AI applications, Retrieval-Augmented Generation (RAG) frameworks, and Agentic AI workflows. Deep proficiency in fine-tuning open-source LLMs (Llama 3, Mistral), vector databases (Pinecone, Qdrant, Milvus), and deploying scalable inference pipelines with vLLM.

TECHNICAL EXPERTISE

GenAI & LLMs
RAG, LangChain, LlamaIndex, Prompt Engineering, LoRA/QLoRA, vLLM, Ollama

Vector Stores & NLP
Pinecone, Qdrant, ChromaDB, Hugging Face Transformers, Tokenization, BERT

Engineering & Cloud
Python, FastAPI, PyTorch, Triton Inference Server, Docker, AWS SageMaker

WORK EXPERIENCE

Lead AI Engineer • **LatentView Analytics**
Jan 2022 – Present | Chennai

- Architected an enterprise RAG query system over 2M confidential financial filings with hybrid semantic search and reranking.
- Fine-tuned open-source LLMs using LoRA/QLoRA on specialized legal corpora, boosting extraction accuracy from 71% to 92.4%.
- Accelerated model serving throughput by 3.8× using vLLM continuous batching and PagedAttention techniques on NVIDIA A100 GPUs.

AI / Machine Learning Engineer • **Birlasoft**
Aug 2019 – Dec 2021 | Coimbatore

- Constructed real-time conversational assistants using Rasa and transformer models handling 40,000 monthly user inquiries.
- Built automated text summarization and named entity recognition (NER) pipelines with SpaCy and Hugging Face.
- Containerized AI scoring microservices with FastAPI and Docker on AWS ECS clusters.

FEATURED AI PROJECTS

Autonomous Multi-Agent Market Research System
Engineered an agentic framework using LangGraph coordinating 4 specialized agents to browse, parse, critique, and synthesize investment reports.

Sub-Second Semantic Search Engine
Deployed Qdrant vector cluster indexing 10M embeddings with HNSW graphs, delivering sub-30ms semantic search across e-commerce catalogs.

EDUCATION & PUBLICATIONS

M.Tech Artificial Intelligence • Amrita Vishwa Vidyapeetham, Coimbatore

Published Paper: "Optimized Context Compression for Domain-Specific RAG" (IEEE 2023)